

Timely Student Support

From fragmented signals to governed action on Databricks •
Synthetic demonstration



Advisors need to know who
needs attention today—and
why—without compromising
trust.

Buyer KPIs and synthetic demo baseline

88.88%

Synthetic retention rate

2,664

Initially elevated-risk students

\$24.1M

Estimated synthetic tuition exposure

20,000

Current synthetic students

Current-state friction delays action

Fragmented signals

Attendance, LMS, finance, enrollment, and case data arrive on different clocks.

Inconsistent prioritization

Advisors reconcile data manually and lack a defensible daily queue.

Weak feedback loop

Outreach and outcomes are disconnected from executive coverage and analytics.

A closed-loop workflow from signal to action

1. Observe — trigger one deterministic synthetic day
2. Govern — apply quality, lineage, grants, and row filters
3. Prioritize — score current snapshots with a versioned ML model
4. Assist — generate cited, evaluated advisor briefings
5. Act — record interventions transactionally in Lakebase
6. Learn — return CDC to governed Gold
7. Understand — ask embedded Genie and inspect its SQL

Business value for two accountable owners

Executive sponsor

See aggregate risk concentration, intervention coverage, retention, and estimated tuition exposure—without student detail.

Student-success owner

Reduce signal-to-outreach time with a prioritized caseload, grounded context, and closed-loop follow-up history.

Governed Databricks architecture

Governed intelligence

Lakeflow Bronze + Silver

MLflow scoring

Unity Catalog Gold

Unified experience

Databricks App

Read-only Lakebase serving

Embedded Genie Agent

Closed feedback loop

Lakebase-owned interventions

CDC to governed Gold

Serving cannot overwrite writes

Hybrid intelligence keeps each decision explainable

ML classifier

Repeatable prioritization, calibrated scores, stable tiers, version history, and baseline comparison.

Grounded GenAI

Allowlisted facts, citations, bounded discussion prompts, and fixed-set evaluation.

Human advisor

Judgment, outreach, reconciliation, and accountable action remain with people.

Live demo: one coherent student-success story

Signal → action

Replay the synthetic incident day

Inspect quality and feature freshness

Show a medium-to-high risk transition

Review the cited advisor briefing

Action → learning

Record outreach and follow-up

Verify CDC and intervention Gold

View aggregate executive coverage

Ask embedded Genie and inspect its SQL

Execution evidence—not screenshots

60,000

Immutable daily risk-history rows

0.9669

Synthetic validation PR-AUC

SUCCESS

Full workflow and idempotent replay

ONLINE

Both serving sync tables

Trust is designed in—and limitations stay visible

Trust controls

Row-filtered advisor access

Aggregate-only executive products

Read-only serving tables

Cited AI and inspectable SQL

Visible stale, partial, and error states

Honest boundaries

Synthetic data and designed signals

No causal impact claim

No automated adverse action

No institutional validation yet

Triggered daily—not real time

